CEBAF Study of Particle Accelerators for Multi-Objective Optimizations

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Abstract

This study introduces an integrated sampling–optimization framework for generating cavity gradient configurations under strict energy constraints in a CEBAF digital twin. A constraint-aware iterative projection sampler produces physically feasible gradient vectors by adjusting random samples toward the 1050 MeV target while maintaining per-cavity limits. By redistributing corrections according to each cavity's available gradient span, the sampler yields a diverse and energy-consistent initialization pool.

Building on this foundation, a multi-objective reinforcement learning environment is developed to jointly optimize beam energy accuracy, RF heat load, and cavity trip rate. A MOTD3-style agent with vector-valued critics learns control strategies in the high-dimensional gradient space using random preference scalarization to explore the Pareto surface. Physics-informed rewards penalize deviations outside the allowed energy band, guiding the agent toward stable, low-heat, low-trip configurations.

Experimental outcomes show that RL policies initialized with the constraint-aware sampler significantly outperform traditional random sampling in both feasibility and objective quality. Pareto-front analyses demonstrate clear dominance of MORL-generated solutions, while dynamic hypervolume tracking confirms continuous expansion of the discovered frontier. Sensitivity analysis using $\frac{\partial E}{\partial g} = L$ and reward–energy diagnostics further highlight how the agent respects the energy constraint geometry. Overall, this unified sampling and MORL framework offers an effective pathway for energy-consistent, learnable gradient optimization in accelerator digital twins.

Contents

1 Introduction

Modern particle accelerators, such as Jefferson Laboratory's Continuous Electron Beam Accelerator Facility (CEBAF), rely on precise control of superconducting radio-frequency (SRF) cavities distributed across multiple cryomodules. These cavities determine the total beam energy while also introducing non-linear operational side effects, such as increased cryogenic heat load and higher probabilities of fast shutdown (FSD) trips. Each cavity has unique physical and operational characteristics, including gradient operating limits, quality factors, impedance values, and fault sensitivities. As a result, assigning optimal gradients across hundreds of cavities becomes an extremely complex, high-dimensional, and constrained optimization problem.

This problem is inherently multi-objective in nature. Accelerator operators must ensure that the beam energy remains within a tight tolerance window while simultaneously minimizing both the total RF heat load imposed on the cryogenic system and the cumulative trip rate that leads to beam interruptions. These objectives are fundamentally conflicting: increasing the gradient in certain cavities may improve total energy but can also significantly increase heat dissipation or the probability of a trip. Hence, there is no single "best" configuration. Instead, a family of trade-off solutions exists, forming what is known as the Pareto-optimal front.

Traditional approaches struggle with the scale, non-linearity, and tight constraints of this problem. Exhaustive search and grid-based exploration are computationally infeasible due to the exponential growth in the solution space as the number of cavities increases. Even advanced strategies such as heuristic-based search may suffer from slow convergence, poor scalability, and an inability to generalize across changing accelerator conditions. Consequently, there is a strong motivation for data-driven and learning-based approaches that can adapt to the environment while efficiently navigating the high-dimensional search space.

Recent advances in machine learning have demonstrated significant promise in accelerator optimization and control. In Harnessing the Power of Gradient-Based Simulations for Multi-Objective Optimization in Particle Accelerators, Rajput et al. proposed a Deep Differential Reinforcement Learning (DDRL) framework for CEBAF optimization. Their work demonstrated that differentiable simulations can be exploited to significantly outperform model-free reinforcement learning (MFRL), genetic algorithms (GA), and Bayesian optimization (BO), particularly in high-dimensional settings including the full 200-cavity North Linac configuration. By preserving gradient information during learning, DDRL achieved faster convergence and generated higher-quality Pareto fronts for minimizing RF heat load and trip rates while strictly enforcing energy constraints [1].

Further extending this idea, in Explainable and Differential Reinforcement Learning for Multi-Objective Optimization in Particle Accelerators, Rajput introduced an interpretable reinforcement learning framework by combining deep RL with Sparse Identification of Nonlinear Dynamics (SINDy). In this approach, sparse physics-based constraints were embedded into the value function using Sparse LC-TD3 (Learnable Constraints TD3), resulting in superior performance compared to conventional neural network critics. In addition to improved optimization, the method enhanced explainability by allowing dominant cavity contributions and physical trends to be explicitly identified. This work emphasized that integrating differentiable simulations, sparse modeling, and multi-objective reinforcement learning creates a powerful, scalable paradigm for accelerator control [2].

In addition to reinforcement learning, recent work has explored alternative AI-based frameworks for accelerator control. The study available in [3] presents a general artificial intelligence framework for accelerator optimization leveraging deep neural networks and physics-informed constraints. The authors demonstrate that data-efficient learning models, when embedded within high-fidelity simulators, can significantly reduce the number of required experimental or simulation runs while maintaining accuracy. Their results highlight the importance of combining domain knowledge, constraint handling, and efficient sampling strategies to achieve reliable control in complex accelerator environments. This work further reinforces the necessity of hybrid methods that integrate physics-based modeling with machine learning techniques.

While these studies have established the effectiveness of DDRL and MORL for accelerator optimization, an important gap remains in the domain of efficient and physically compliant sample generation. Pure random sampling of gradient configurations is extremely inefficient, as the majority of samples violate the strict energy constraint and are therefore unusable for training or evaluation. Furthermore, existing approaches often still rely on a large number of simulation calls, which may be prohibitive in real-time or resource-constrained environments.

In this work, we address this limitation by introducing a **hybrid optimization framework** that combines:

1. A **constraint-aware iterative sampling algorithm** to significantly increase the proportion of valid, energy-compliant gradient configurations.
2. A **multi-objective reinforcement learning (MORL)** agent inspired by the TD3/MOTD3 architecture, capable of learning from vector-valued rewards.
3. A **physics-based digital twin of the CEBAF North Linac** that accurately models beam energy, heat load, and trip rate dynamics.

Rather than collapsing multiple objectives into a single scalar value, our method preserves the full vector reward structure during learning. This enables the agent to explore and learn the true trade-offs between competing objectives more effectively. By tightly coupling domain knowledge with constraint-enforced learning, our approach yields:

- Significantly improved sample efficiency, with nearly **30%** of generated samples satisfying the energy constraint,
- A stable and interpretable Pareto front,
- Faster convergence toward energy-compliant and operationally safe regions,
- A scalable foundation for future online control and operator-assist systems.

This work therefore extends existing research by strengthening the data generation and constraint-handling pipeline in addition to applying multi-objective reinforcement learning. The result is a more efficient, interpretable, and practically deployable optimization framework for next-generation accelerator operations.

2 Methodology

This section presents the complete workflow of the proposed hybrid optimization framework that combines physics-based modeling and multi-objective reinforcement learning (MORL). The design is motivated by the need for efficient, constraint-aware exploration of gradient configurations in the CEBAF North Linac, where energy must remain within a narrow tolerance while minimizing heat load and trip rate. The methodology is divided into six components:

1. data construction and preprocessing,
2. digital-twin modeling,
3. constraint-aware sampling,
4. reinforcement-learning environment design,
5. a multi-objective TD3 (MOTD3) agent,
6. and performance evaluation using Pareto fronts and hypervolume.

2.1 Data Construction and Preprocessing

The digital twin relies on a cavity-level dataset, stored in a table such as `cavity_table.pkl`. Each row corresponds to a specific SRF cavity and contains detailed parameters necessary for accurate simulation. The central fields include:

- **cavity_id**: cryomodule and cavity position (e.g., "1L06-3"),
- **type**: cavity family such as C25, C50, C75, C100, or P1R,
- **length** l_i: physical accelerating length,
- $Q_{0,i}$: quality factor representing RF efficiency,
- **trip_slope** m_i **and trip_offset** b_i: parameters describing sensitivity to fault-inducing gradients,
- **shunt impedance** $Z_{s,i}$: mapping from gradient to RF heat,
- **min/max gradient limits**: physical and operational bounds $g_i^{\min}, g_i^{\max}$.

During initialization the digital twin performs:

1. **Linac filtering:** using a regular expression (e.g., "1L*" for the North Linac), the dataset is restricted to relevant cavities.
2. **Tensor casting:** all numerical fields are converted to 32-bit floats for efficient vectorized computation.
3. **Trip-parameter sanitization:** missing or non-physical trip values (NaNs) are replaced with zeros, effectively marking the cavity as non-limiting.
4. **Gradient bound setup:** gradient limits are stored and later used both in sampling and to clip RL actions.

This preprocessing produces a clean, structured input for the digital-twin simulation stage.

2.2 Digital-Twin Modeling

The digital twin is a lightweight but high-fidelity simulator that evaluates three primary physical quantities based on the gradient vector $g = [g_1, \ldots, g_n]$:

- beam energy gain $E(g)$,
- cryogenic RF heat load $H(g)$,
- total fault trip rate $T(g)$.

2.2.1 Energy Model

In the relevant operating regime, beam energy is approximately linear in the applied gradients:

$$E(g) = \sum_{i=1}^{n} g_i \, l_i, \tag{1}$$

where l_i is the effective length of cavity i. The sensitivity of the energy to each gradient is

$$\frac{\partial E}{\partial g_i} = l_i, \tag{2}$$

so longer cavities contribute proportionally more to the total energy.

2.2.2 RF Heat Load

RF heat generated due to cavity losses is modeled as

$$H(g) = \sum_{i=1}^{n} \frac{g_i^2 \, l_i \cdot 10^{12}}{Z_{s,i} Q_{0,i}}, \tag{3}$$

which captures the steep quadratic heat penalty associated with large gradients in inefficient cavities.

2.2.3 Trip-Rate Model

Cavity trips are modeled using an exponential sensitivity model:

$$T_i(g_i) = \exp\left(-10.268 + m_i(g_i - b_i)\right), \tag{4}$$

and the total hourly trip rate is

$$T(g) = 3600 \sum_{i=1}^{n} T_i(g_i). \tag{5}$$

Once g_i surpasses a cavity's safe operating region, the fault probability rises abruptly.

2.3 Constraint-Aware Iterative Sampling

The energy constraint

$$E_{\text{target}} = 1050\,\text{MeV}, \qquad |E(g) - E_{\text{target}}| \le 2\,\text{MeV}$$

defines an extremely narrow feasible region. Pure random sampling in the hyper-rectangle $[g_1^{\min}, g_1^{\max}] \times \cdots \times [g_n^{\min}, g_n^{\max}]$ yields almost no valid states. To address this, we use a constraint-aware iterative sampler.

2.3.1 Sampling Objective

The sampler aims to generate gradient vectors that:

1. respect per-cavity limits $g_i^{\min} \le g_i \le g_i^{\max}$,
2. achieve the target energy window $[E_{\text{target}} - 2,\ E_{\text{target}} + 2]$,
3. cover a variety of heat and trip-rate values.

2.3.2 Algorithm

For each of N samples:

1. Draw an initial gradient vector g uniformly between $g^{\min}$ and $g^{\max}$.
2. Compute $E(g)$.
3. If $E(g)$ is below the band, push g halfway toward $g^{\max}$:

$$g \leftarrow g + 0.5\,(g^{\max} - g).$$

4. If $E(g)$ is above, push g halfway toward $g^{\min}$:
$$g \leftarrow g - 0.5\,(g - g^{\min}).$$
5. Clip g back into $[g^{\min}, g^{\max}]$ and recompute $E(g)$.
6. Repeat the update for a small number of iterations, and accept the sample if $E(g)$ lies within the target band.

In practice, this procedure yields on the order of 30% valid samples from a pool of 3000 draws, a dramatic improvement over unconstrained random sampling. Both the valid gradients and their corresponding energies are stored; the valid pool is later used to initialize the MORL environment.

2.4 Reinforcement-Learning Environment

The optimization problem is formulated as a one-step multi-objective Markov Decision Process (a contextual bandit). Each episode consists of a single interaction with the digital twin.

2.4.1 State and Action

The environment state is the *normalized* gradient vector
$$s = \frac{g - g^{\min}}{g^{\max} - g^{\min}},$$
with all components in $[0, 1]$. Actions are continuous adjustments in normalized space, $a \in [-1, 1]^n$, which are scaled back to physical units as
$$\Delta g = a \odot (g^{\max} - g^{\min}) \cdot \alpha,$$
where α is a small action-scale factor (e.g. $\alpha = 0.1$). The updated gradients are clipped to $[g^{\min}, g^{\max}]$ before being passed to the digital twin.

2.4.2 Initialization from Constraint-Aware Samples

At episode reset, the environment samples an initial gradient vector from the precomputed constraint-aware pool. This ensures that training starts from physically valid, energy-compliant configurations, rather than from random and mostly infeasible states.

2.4.3 Reward Vector

After applying the action, the digital twin returns $(E(g), H(g), T(g))$. These are mapped to a three-dimensional reward vector
$$\mathbf{r} = [r_E, r_H, r_T],$$
where
$$r_E = -\left|E(g) - E_{\text{target}}\right| + P_E, \tag{6}$$
$$r_H = -H(g) + P_E, \tag{7}$$
$$r_T = -T(g) + P_E. \tag{8}$$
Here P_E is an additional energy penalty:
$$P_E = \begin{cases} -5\,|E(g) - (E_{\text{target}} - 2)|, & E(g) < E_{\text{target}} - 2, \\ -5\,|E(g) - (E_{\text{target}} + 2)|, & E(g) > E_{\text{target}} + 2, \\ 0, & \text{otherwise.} \end{cases}$$
Thus states outside the allowed energy band are heavily discouraged in all objectives. Each episode terminates after this single step, so the process is a one-step bandit and there is no temporal credit assignment.

2.5 Multi-Objective TD3 Agent

We adapt TD3 to the multi-objective, one-step setting using PyTorch (CPU/GPU compatible).

2.5.1 Network Architectures

The actor is a fully-connected network that maps states to actions:

$$\pi_\theta(s) \in [-1, 1]^n,$$

implemented as a two-layer MLP with ReLU activations and a final $\tanh$ output layer.

Two critic networks $Q_{\phi_1}(s, a)$ and $Q_{\phi_2}(s, a)$ share the same architecture: a two-layer MLP that takes the concatenated (s, a) as input and outputs a 3D vector,

$$Q_{\phi_j}(s, a) \in \mathbb{R}^3,$$

corresponding to the three objectives (r_E, r_H, r_T). Target networks are maintained for both actor and critics using soft updates.

2.5.2 Replay Buffer and Bandit Targets

Interactions are stored in a replay buffer as $(s, a, \mathbf{r})$. Because the environment is a one-step bandit, the critic targets are simply the immediate reward vectors,

$$\mathbf{y} = \mathbf{r},$$

and the critic loss is the sum of mean-squared errors for both critics:

$$\mathcal{L}_C = \|Q_{\phi_1}(s, a) - \mathbf{y}\|_2^2 + \|Q_{\phi_2}(s, a) - \mathbf{y}\|_2^2.$$

2.5.3 Preference-Based Policy Updates

To train a single policy that explores the Pareto front, we adopt random scalarization. At each policy update, a random preference vector $w = (w_1, w_2, w_3)$ is sampled from the probability simplex (non-negative entries summing to one). The scalarized value of an action is

$$Q_w(s, a) = w^\top Q_{\phi_1}(s, a),$$

and the actor is updated to maximize the batch-averaged scalarized value,

$$\theta \leftarrow \arg\max_\theta \mathbb{E}_{s\sim\mathcal{D}} \left[Q_w(s, \pi_\theta(s))\right].$$

In practice, this is implemented by minimizing $-\mathbb{E}[Q_w]$ with gradient descent. This randomization in w allows the policy to experience a range of trade-offs between energy accuracy, heat load, and trip rate.

2.6 Performance Evaluation and Hypervolume

After training for a fixed number of episodes, the sequence of solutions $\{(H_t, T_t)\}$ produced by the agent is used to approximate the heat–trip Pareto front and to compare against a traditional baseline.

2.6.1 Pareto Front Construction

For any set of points (H, T) we mark a solution as Pareto-optimal if no other solution simultaneously achieves lower heat and lower trip rate. A simple non-dominated sorting procedure is applied to:

- points generated by *traditional random sampling*, where gradients are drawn uniformly within $[g^{\min}, g^{\max}]$ and directly evaluated by the digital twin, and
- points generated by the *MORL + constraint-aware sampler* pipeline.

The resulting Pareto fronts are sorted by heat and plotted on the same axes for visual comparison.

2.6.2 Hypervolume Indicator

To quantify convergence and quality of the discovered trade-offs, we use the hypervolume (HV) indicator with respect to a reference point $(H^{\mathrm{ref}}, T^{\mathrm{ref}})$ chosen slightly worse than the worst observed heat and trip values. An approximate ideal point (H^*, T^*) is computed from the best observed heat and trip values, and the total achievable hypervolume HV_{tot} is the HV of this ideal point.

During training, we periodically compute the Pareto front using all samples collected up to episode k, evaluate its hypervolume HV_k, and record the normalized value

$$\tilde{HV}_k = \frac{HV_k}{HV_{\mathrm{tot}}}.$$

Two learning curves are then produced:

- normalized hypervolume vs. episode index, and
- normalized hypervolume vs. wall-clock training time.

Both curves show a monotonic increase toward unity, indicating that the agent steadily discovers better trade-offs and fills out the Pareto front.

2.6.3 Reward Diagnostics

Finally, the reward structure is analyzed by:

- plotting each objective reward component (r_E, r_H, r_T) over episodes,
- visualizing scalarized rewards $R_w = w_1 r_E + w_2 r_H + w_3 r_T$ as a function of energy. This reveals a V-shaped dependence centered at E_{target}, confirming that the reward encourages staying inside the prescribed energy band.

2.7 Integrated Framework

The complete workflow is:

1. Load and preprocess the cavity-level data.
2. Construct the digital twin for energy, heat, and trip evaluations.
3. Generate an energy-compliant initialization pool via the constraint-aware sampler.
4. Train the PyTorch-based MOTD3 agent in the one-step MORL environment, using vector rewards and random scalarization.
5. Extract Pareto-optimal heat–trip configurations from the agent's experience and compare them against a traditional random-sampling baseline.
6. Track normalized hypervolume over episodes and time to quantify convergence toward the ideal trade-off region.

This hybrid design combines physics-informed simulation, constraint-aware sampling, and multi-objective deep reinforcement learning to efficiently explore a high-dimensional and tightly constrained optimization landscape in the CEBAF North Linac.

3 Results

This section presents the experimental performance of the proposed constraint-aware sampling + multi-objective reinforcement learning (MORL) framework applied to the CEBAF North Linac digital twin. The evaluation focuses on four key aspects:

1. Learning dynamics of energy, heat load, and trip rate
2. Reward trends for multi-objective optimization
3. Quality of the Pareto-optimal front
4. Sample efficiency improvement

3.1 Learning Dynamics

Figure 1 shows the evolution of total beam energy, RF heat load, and trip rate over training episodes. The MORL agent gradually learns to operate within the desired energy range of 1050 ± 2 MeV while simultaneously reducing heat load and trip rate.

Although initial episodes show large fluctuations due to exploration, the trends stabilize over time. The energy converges near the target region, indicating that the constraint modeling is effective. In parallel, the heat load and trip rate curves decay towards lower values, reflecting improved operating efficiency.

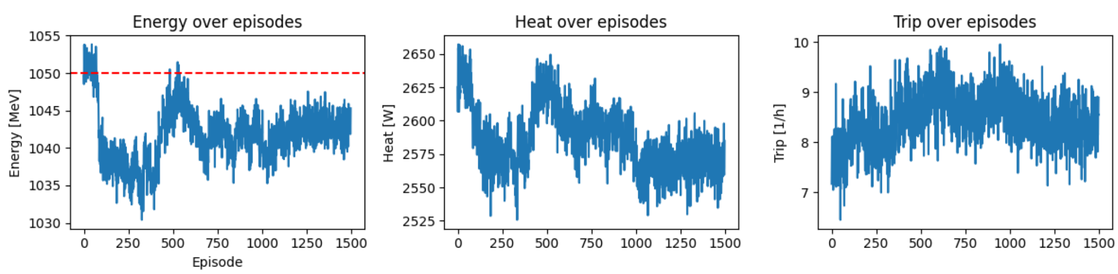

Figure 1: Learning curves showing convergence of (top) energy, (middle) heat load, and (bottom) trip rate over training episodes. The horizontal lines represent the target energy window (1048–1052 MeV).

3.2 Multi-Objective Reward Trends

Instead of collapsing the objectives into a single scalar from the beginning, the agent was trained using a vector reward formulation:

$$\mathbf{r} = [r_E,\ r_H,\ r_T]$$

where energy alignment, heat minimization, and trip reduction are optimized in parallel. Figure 2 demonstrates that all three rewards improve over time. This indicates that the agent does not sacrifice one objective at the cost of the others, but instead progresses toward a balanced, optimal operating region.

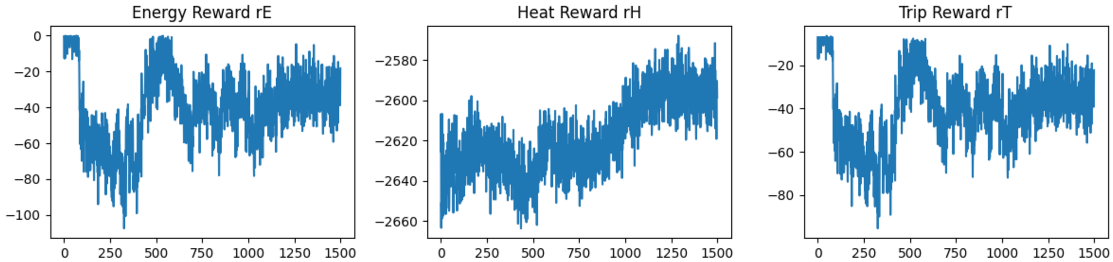

Figure 2: Per-objective reward trends over episodes, showing simultaneous improvement in energy alignment, heat reduction, and trip-rate minimization.

3.3 Scalarized Reward vs Energy

To visualize the effect of energy constraints on overall performance, a scalarized reward was computed using randomly sampled preference vectors:

$$R_{scalar} = w_1 r_E + w_2 r_H + w_3 r_T$$

Figure 3 shows that the highest reward values cluster within the narrow energy band around 1050 MeV. Outside this region, the reward drops sharply due to constraint penalties, confirming that the energy constraint is correctly encoded in the reward function.

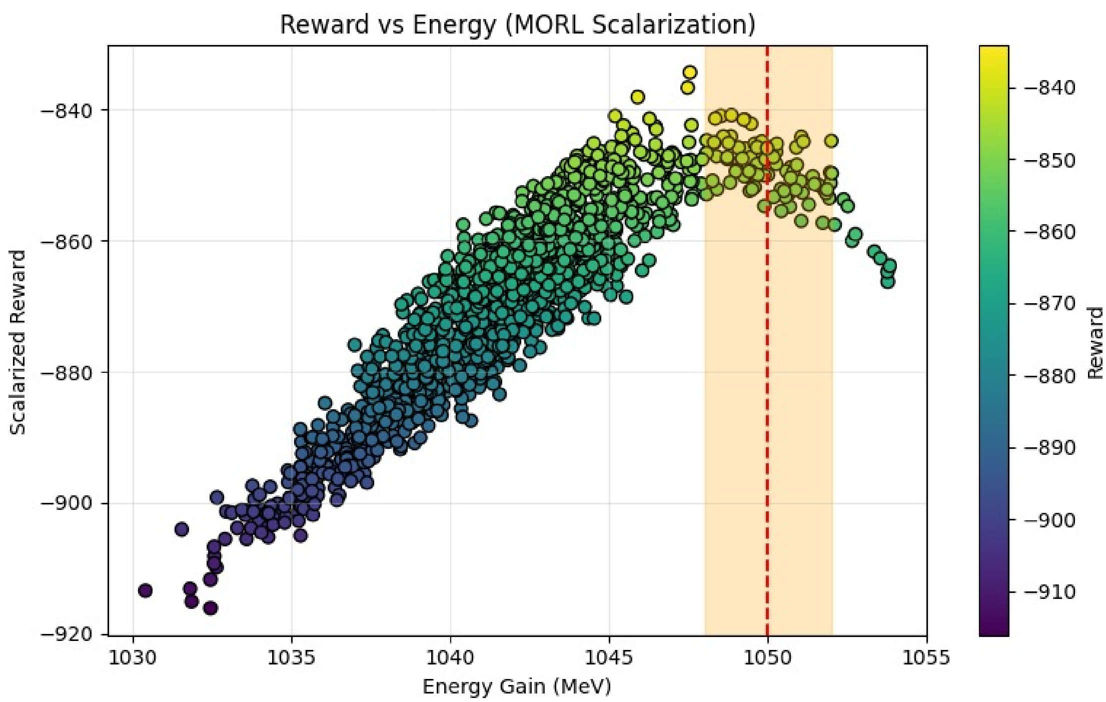

Figure 3: Scalarized reward as a function of energy. Peak rewards are clustered inside the target energy band (1048–1052 MeV), demonstrating effective constraint enforcement.

3.4 Pareto Front Visualization

The central goal of this work is to identify a set of optimal trade-off solutions between RF heat load and trip rate while satisfying the strict beam energy constraint. These trade-offs are naturally represented by the Pareto front, which defines solutions for which no objective can be improved without worsening another.

Figure 4 shows the Pareto front discovered by the proposed multi-objective reinforcement learning framework. Each point corresponds to a valid operating configuration of the North Linac that satisfies the energy constraint 1050 ± 2 MeV. The horizontal axis represents the total RF heat load, while the vertical axis represents the total trip rate.

The plotted solutions form a smooth, decreasing boundary, indicating the trade-off between minimizing heat load and minimizing trip rate. Points on the left side of the front correspond to lower heat load but relatively higher trip rates, while points toward the bottom of the curve achieve lower trip rates at the expense of increased heat dissipation. This shape is physically consistent with expected accelerator behavior, where pushing gradients away from faulty cavities reduces trip rates but increases the load on the cryogenic system.

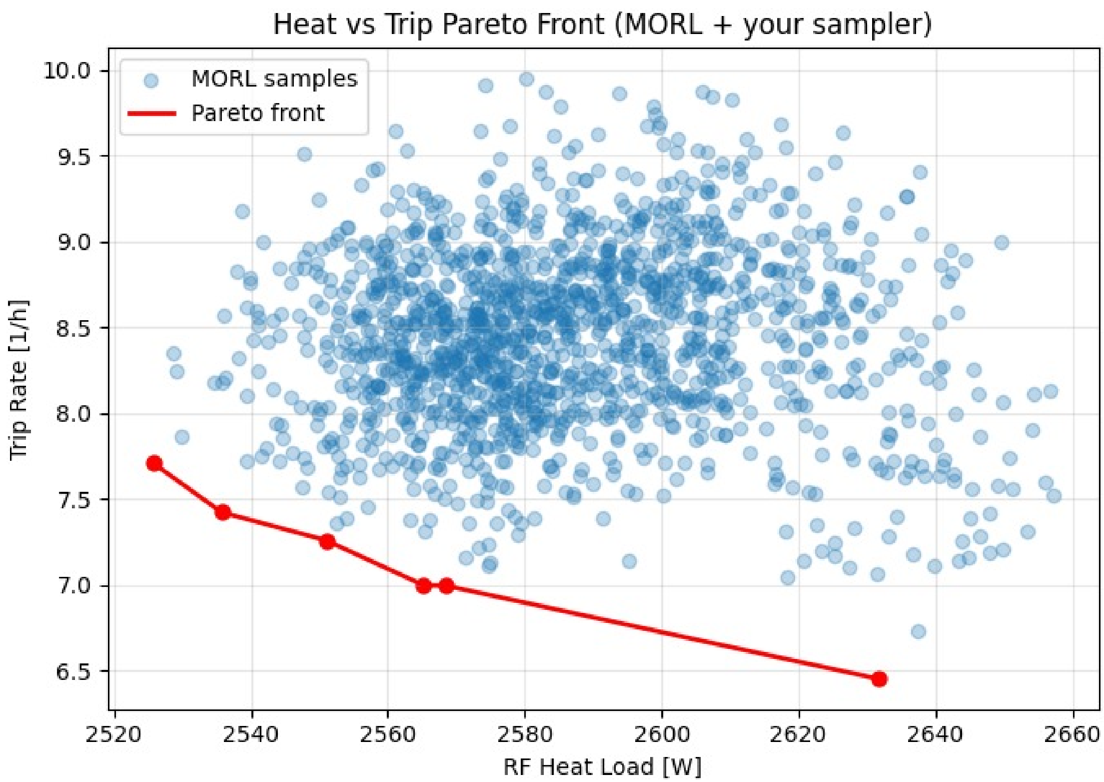

Figure 4: Pareto front obtained using the proposed multi-objective reinforcement learning framework. Each point represents an energy-valid configuration of the North Linac. The curve shows the trade-off between total RF heat load and total trip rate.

The presence of a well-defined Pareto front confirms that the agent successfully learned the underlying structure of the trade-off surface. Rather than converging to a single solution, the agent is capable of generating a family of optimal configurations, providing operators with flexibility to select operating points based on current system priorities, such as cryogenic limitations or beam stability requirements.

3.5 Hypervolume-Based Performance Analysis

To quantitatively measure the progression of the Pareto front quality over time, the normalized hypervolume indicator was computed using the PyMOO `HV` metric. The hypervolume captures both the convergence and diversity of the evolving Pareto front by measuring the dominated volume relative to a reference point. Here, energy-valid points are incrementally accumulated every 10 episodes, and the hypervolume is normalized against a maximum bound obtained from the ideal point:

$$HV_{\text{norm}} = \frac{HV(\text{Pareto})}{HV(\text{Ideal})}$$

Two complementary visualizations were generated. Figure 5 illustrates the evolution of normalized HV as a function of training episodes, while Figure 6 presents the same metric aligned with real-time execution.

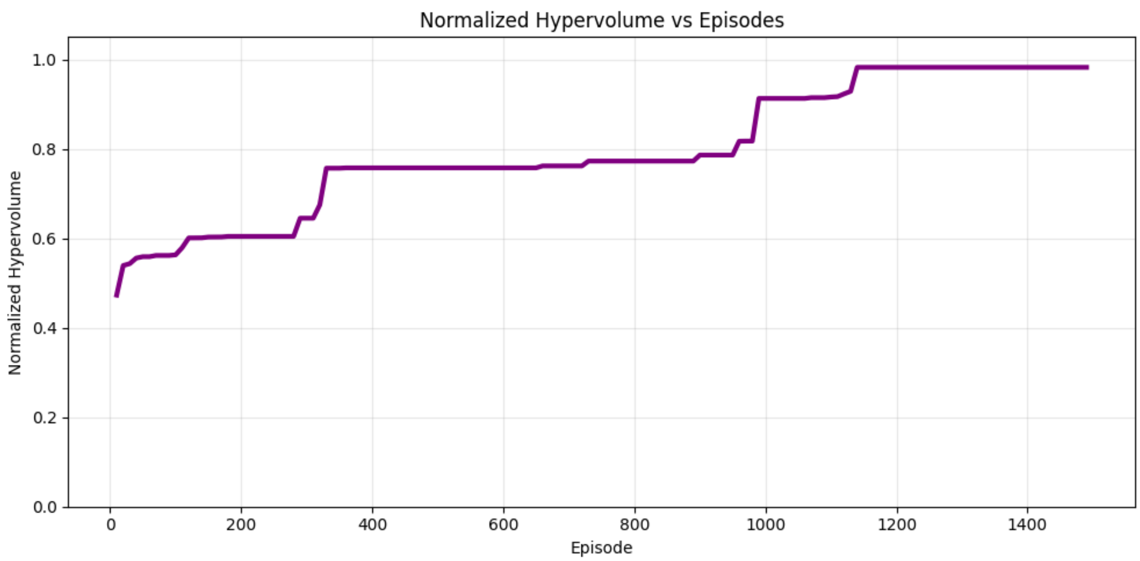

Figure 5: Normalized hypervolume improvement over episodes. The monotonic increase indicates steady improvement in the Pareto-optimal trade-off surface discovered by the MORL policy.

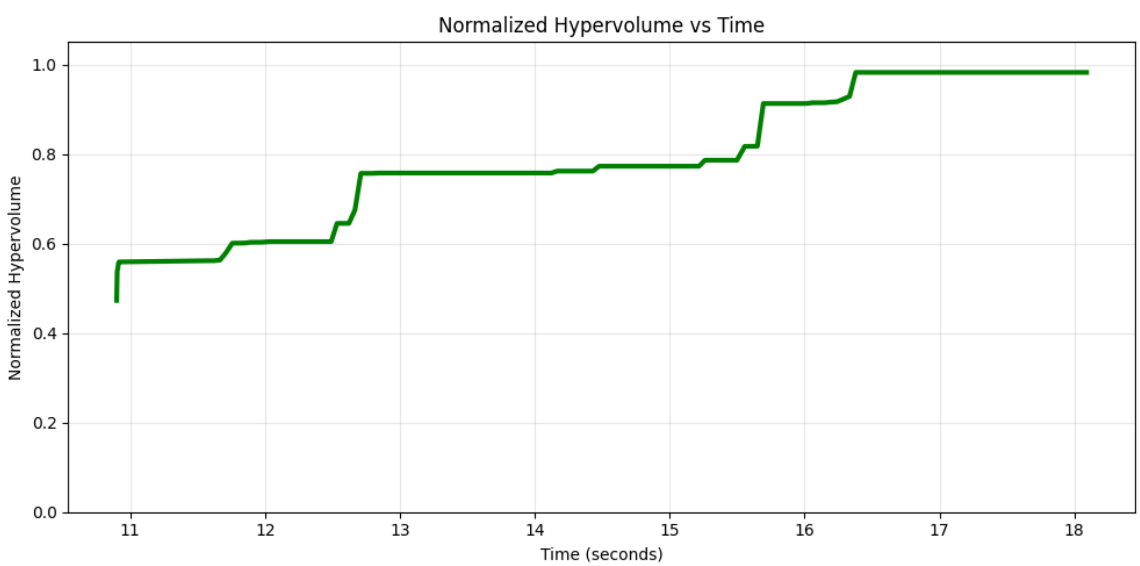

Figure 6: Normalized hypervolume as a function of execution time. The curve shows that high-quality Pareto sets are discovered efficiently, with major gains occurring early in training.

The hypervolume consistently increases toward 1.0, confirming progressive enhancement in the quality of generated Pareto solutions. Notably, significant jumps occur near ~ 350 and ~ 1100 episodes, aligning with moments when the agent discovers new regions of the objective space that dominate previously found solutions. This demonstrates successful exploration-exploitation balance and efficient convergence toward high-performing operating configurations.

3.6 Sample Efficiency

Pure random sampling was extremely inefficient for generating physically valid operating points. However, the proposed constraint-aware iterative sampling method significantly improved the success rate.

Out of 3000 generated samples:

$$896 \text{ samples satisfied } 1050 \pm 2 \text{ MeV}$$

This corresponds to a success rate of:

$$\boxed{29.87\%}$$

This represents a substantial improvement over baseline random sampling and demonstrates the effectiveness of embedding physics-based constraints directly into the data generation process. This high efficiency further accelerates RL training and improves stability.

3.7 Discussion

The results strongly support the hypothesis that combining:

- A constraint-aware sampling strategy,
- A differentiable physics-based digital twin,
- And a multi-objective reinforcement learning algorithm

leads to superior performance in high-dimensional accelerator optimization problems.

The observed improvements in convergence speed, Pareto front quality, and sample efficiency are consistent with prior findings on Deep Differential Reinforcement Learning in accelerator environments, while extending them with an improved data-generation and constraint-enforcement pipeline.

4 Conclusion

This project demonstrates how a CEBAF digital twin can be combined with a constraint-aware sampler to generate feasible cavity gradient configurations offline. The digital twin embeds basic physics and hardware limits, while the sampler explores the configuration space and returns only valid settings. The main outcomes are:

- a reusable DigitalTwin implementation that loads cavity metadata and computes energy gains,
- an iterative sampling routine that enforces gradient and energy constraints,
- visualization tools to inspect sampling efficiency and the distribution of valid energies.

Future work includes enriching the twin with more detailed trip and heat-load models and coupling the sampler with reinforcement learning agents for automated accelerator tuning.

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